



Michael Tarlton

Computational Science for
AI and Systems Neuroscience



Oslo, Norway



+47 41 33 48 76



M@Tarlton.info
Michaelt@oslomet.no



www.Tarlton.info

Professional Summary

A decade plus of experience in computational scientific research and technology related fields, with a background in computational mathematics, statistical physics, and neuroscience. I have a keen focus on studying the mechanisms of the human brain, and the fundamentals of distributed information. I'm pursuing the emerging field of neuroscience based AI models and AI-specific design for low-energy edge hardware. My aim is to create novel learning methods for spiking neural networks and physics-based models for the next generation of machine learning.

Professional Experience

PhD Candidate

Oslo Metropolitan University,
Faculty of Art and Technology - AI Lab
April 2022 – Present

PhD candidate in artificial intelligence with a focus in deep reinforcement learning for spiking neural networks project. Researching novel methods of learning for deep neural networks using time-based communication for application in low-energy edge hardware.

Genetics Data Researcher

Norwegian Centre for Mental Disorders
Research, University of Oslo
January 2022 – April 2022

Applied statistical genomic research using data from Genome-Wide Association Studies. Secure health data server services.



Masters of Science in Neuroscience

Kavli Institute - Roudi Group
Norwegian University of Science and
Technology
September 2018 – April 2021

Evaluated a novel information-based Bayesian criterion for network topological inference from spiking neural activity. Designed and simulated spiking networks with Ising model methods. Compared multiple network topologies for informational transfer and inferability via Bayesian methods.

ML Data Scientist

Atlas Wearables
(acquired by Peloton Interactive)
December 2017 – July 2018

Researcher and data analyst for Atlas “Motion Genome” project. Applied animal biometrics and movement data for proprietary machine learning algorithms.

Research Assistant

xREZ Art + Science Lab
University of North Texas
December 2017 – July 2018

Astrophysics consultant for a multidisciplinary team, together creating mixed media installations meant to convey scientific concepts. Managed astronomical object data collection, analysis, and scientific contextualization for the *Instrument: One Antarctic Night* installation.



Bachelors of Science in Physics Bachelors of Arts in Mathematics

University of North Texas
September 2012 – May 2017

Focus in scientific computing, information theory, statistical astrophysics. Undergraduate thesis studied prominent active galactic nuclei photometric properties via statistical analysis of large sky survey data repositories.

Skills

Scientific

- Computational Statistical Physics and Mathematics
- Neuroscientific Modeling
- Machine Learning Models
- Neuromorphic Hardware Design
- Spiking Neural Networks
- Information Theory
- Complex Dynamical Systems

Technology

- Python
 - PyTorch, Pandas, Tensorflow
- AI Experiment Tracking
 - AimStack
- Git Pipelines and Deployment
- HPC and Cloud Compute Environments
- Docker & Linux Environments
- MATLAB
- SysVerilog

Practical

- Research Methods
- Experimental Design
- Team Organizing
- Project Management
- Knowledge Management and Distribution

Languages

- English – Native Speaker
- Norwegian – A2

Publications

- Tarlton et al. (2025) "Investigation of Novel SBF-Automata Architecture for Periodicity Finding Solutions at Edge Systems" 2025 12th International IEEE/EMBS Conference on Neural Engineering (NER).
- Tarlton et al. (2024) "Neurological Based Timing Mechanism for Reinforcement Learning." Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems.
- Tarlton, M. (2021), Novel Model Selection Criterion for Inference of Ising Models [Master's thesis, Norwegian University of Science and Technology].
- West et al. (2018), "Experiencing a Slice of the Sky: Immersive Rendering and Sonification of Antarctic Astronomy Data"
- Tarlton, M. (2017), Prominent AGN Emission Line Contributions to Photometric Bands, [Unpublished bachelor's thesis], University of North Texas.